

Q-Pharm: An Open-Source, Quantum-Accelerated Drug-Repurposing Engine with VQE-Refined Molecular Docking

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Abstract. Identifying which of the thousands of already-approved drugs bind a newly emerging disease target is a screening problem that has historically demanded licensed software, docking clusters and expert curation. We present Q-Pharm, an open-source, fully software-only engine that takes a protein structure (by PDB identifier or upload), detects its binding pocket, and screens a curated library of 294 FDA-approved drugs through a four-stage pipeline: (i) pharmacophore-guided rigid-body docking scored by an empirical function in the LUDI tradition; (ii) a quantum-refinement stage in which the strongest polar contact of each top-ranked pose is mapped onto a two-site Hubbard Hamiltonian and solved on four qubits with a variational quantum eigensolver (VQE) using exact parameter-shift gradients; (iii) ADMET profiling (Lipinski, Veber, PAINS/BRENK, ESOL) with a compact machine-learning drug-likeness prior; and (iv) a transparent, fixed-weight composite ranking. The quantum module is validated against the analytic half-filled Hubbard-dimer ground state and exact diagonalization (agreement to $4.8 \times 10^{-12} E_h$), and the complete pipeline is demonstrated end-to-end on the SARS-CoV-2 main protease (PDB 6LU7), where the detected pocket reproduces the experimentally established active site and the top-ranked hits include nucleoside-antiviral pharmacophores with dense hydrogen-bond networks. A full 60-drug screen completes in approximately six minutes on a consumer laptop, runs entirely on simulators and public databases, and is released under the MIT license with a web-based screening console. Q-Pharm is intended as an accessible hypothesis-generation tool and as a reproducible scaffold for hybrid quantum-classical screening research; its outputs are computational estimates, not clinical evidence.

Keywords. drug repurposing; molecular docking; variational quantum eigensolver; Hubbard dimer; ADMET; Qiskit; cheminformatics

1 Introduction

Developing a new drug remains one of the most expensive and slowest endeavours in industry: the conventional figure is 12–15 years and on the order of 2.6 billion US dollars per approved molecule, with roughly nine in ten clinical candidates failing. Drug repurposing—finding new targets for molecules whose human safety profile is already established—short-circuits much of this funnel, and its landmark successes are well documented: zidovudine (a failed cancer candidate) became the first approved AIDS therapy; sildenafil moved from angina to erectile dysfunction and pulmonary arterial hypertension; thalidomide was reborn as a myeloma and leprosy treatment; and remdesivir, an Ebola-virus asset, was repositioned into the first widely authorized therapy for COVID-19 [1, 2].

The bottleneck is systematic *screening*. When a new pathogen emerges, one would like to ask a simple question: which approved molecules bind this target? Answering it computationally requires protein structures, molecular conformer generation, a docking score, absorption and toxicity filters, and the expertise to interpret them—resources concentrated in well-funded laboratories. During the COVID-19 response this gap was visible in real time: the structures of the viral protease and polymerase were published within weeks [3, 4], yet

the computational capacity to screen a pharmacopoeia against them was not.

Three developments make it possible to close this gap on ordinary hardware. First, open-source cheminformatics (RDKit) now covers conformer generation, atom typing and descriptor calculation at production quality. Second, public data services (the RCSB Protein Data Bank [15], PubChem [16]) expose structures and validated molecular representations over plain HTTP. Third, variational quantum eigensolvers [10, 11] provide a practical pattern for adding an explicit electronic-structure correction to an otherwise classical pipeline, a direction actively explored for near-term drug discovery [12, 14].

This paper presents Q-Pharm, an open-source engine that combines these ingredients into a single, reproducible tool. Its contributions are:

- **A complete, free screening pipeline** from PDB identifier to ranked, evidence-linked shortlist, requiring no licensed software and no cluster (Section 3).
- **A quantum-refinement stage** that maps the strongest protein–ligand polar contact onto a two-site Hubbard Hamiltonian, maps the fermionic model to four qubits, and solves it with a VQE driven by exact parameter-shift gradients—a work-

ing hybrid quantum–classical loop rather than a mock-up (Section 3.3).

- **Radical transparency:** every candidate exposes its full score decomposition, per-term scoring components, interaction map, VQE diagnostics (qubit count, Pauli-term count, iteration count, error versus exact diagonalization) and literature links; the composite weights are fixed and published.
- **Validation at two levels:** unit and property tests for every module, including agreement of the quantum module with the analytic Hubbard-dimer ground state; and an end-to-end demonstration on the SARS-CoV-2 main protease (Section 5).
- **A fully open release:** MIT-licensed code, a web screening console with interactive 3D pose visualization, and deterministic data-build scripts that reconstruct the drug library from PubChem.

We state the scope plainly at the outset: Q-Pharm’s scores are computational estimates produced by deliberately simplified models. The system is a hypothesis generator for research use; nothing in it constitutes clinical evidence.

2 Background and related work

Empirical docking scores. Since LUDI [5], empirical scoring functions have modelled binding affinity as a weighted sum of physically interpretable terms—hydrogen bonds, hydrophobic contact area, ionic interactions, van der Waals attraction and a conformational-entropy penalty—and remain the standard for large-library screens where physics-based free-energy methods are too expensive. Q-Pharm adopts this structure with calibrated weights (Section 3.2).

Rule-based and ML ADMET filters. The Lipinski Rule of 5 [6], Veber bioavailability rules [7], PAINS substructure alerts [8] and ESOL solubility estimation [9] form the standard in-silico triage cascade and are all implemented here.

Quantum computing for chemistry. The variational quantum eigensolver [10, 11] estimates molecular ground states on near-term quantum hardware by preparing a parameterized trial state and minimizing its energy classically. Reviews [12] survey the mapping of fermionic Hamiltonians to qubits (Jordan–Wigner [13]) and the role of model Hamiltonians such as the Hubbard dimer as tractable benchmarks; hybrid quantum–classical pipelines for drug discovery have been proposed [14]. Q-Pharm instantiates the pattern in an open, reproducible tool rather than a simulation study.

Open cheminformatics infrastructure. RDKit [18] provides the molecular machinery; 3Dmol.js [17] enables browser-side 3D visualization; the PDB [15] and PubChem [16] provide the data substrate.

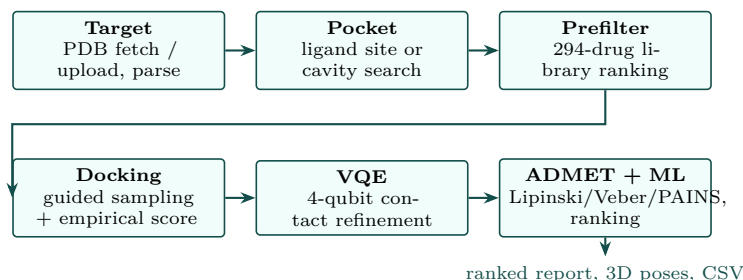


Figure 1: Q-Pharm screening pipeline. Every stage reports progress to the web console; all artefacts (poses, structures, CSV export) are served from the job directory.

3 System design

Q-Pharm is organized as a FastAPI service with a React console. A screening job is a seven-stage pipeline (Figure 1); progress is streamed to the client by polling a job endpoint. The pipeline stages are strictly sequential within a job, and jobs execute on a small worker pool.

3.1 Target acquisition and pocket detection

Structures are fetched from the RCSB PDB over HTTPS or accepted as uploads. The parser keeps the first model, resolves alternate locations, maps modified residues to their parents (e.g. selenomethionine to methionine) and classifies every HETATM group. Pocket definition uses two strategies in order of preference. If a drug-like co-crystal ligand (6–70 heavy atoms, excluding waters, ions, buffers and glycans) is present, the pocket comprises all protein residues with an atom within 6.5 Å of it; for curated targets the reference ligand is pinned explicitly (for example the remediated inhibitor code PJE for the N3 inhibitor in 6LU7). Otherwise, a LIGSITE-style detector samples a 1.4 Å voxel grid, keeps voxels at least 3 Å from any protein atom but inside the protein’s convex hull, and flood-fills connected cavity clusters, selecting the largest.

Both receptor and ligand atoms are typed into four pharmacophore classes: hydrogen-bond donors, hydrogen-bond acceptors, hydrophobic atoms and aromatic ring centroids. Receptor typing uses residue- and atom-name rules; ligand typing uses RDKit SMARTS patterns.

3.2 Docking engine

Conformers. RDKit ETKDGv3 generates 3–6 conformers per drug (adaptive to molecule size), relaxed with MMFF94 (UFF fallback); Gasteiger charges provide ligand partial charges.

Placement. Docking is rigid-receptor with rigid-body sampling. A lattice of *cavity anchors*—points inside the pocket sphere that are themselves at least 3 Å from every receptor atom—guarantees that most sampled placements begin geometrically feasible. Half of the placements receive pharmacophore seeding: one ligand donor or acceptor is placed at the ideal 2.85 Å hydrogen-bond distance from a complementary pocket

atom, with a random orientation about that axis; the remainder use uniform random rotations about anchors. A KD-tree clash filter against the full protein rejects poses with any atom pair below 2.35 Å.

Scoring. Poses are evaluated with an empirical function whose terms and weights are

$$E = 8.0 \sum_{d_i < 2.55} (2.55 - d_i)^2 - 0.062 \sum e^{-(d-3.8)^2/0.8} - 2.6 \sum_{D \cdots A} e^{-(d-2.85)^2/0.2} - 0.105 \sum_{hh} e^{-(d-4.0)^2/0.9} + 83 \sum_{i,j} \frac{q_i q_j}{4d_{ij}^2} - 1.2 N_{\pi\text{-stack}} + 0.35 N_{\text{rot}}, \quad (1)$$

where the first term penalizes steric clashes, followed by van der Waals attraction, hydrogen bonds (donor-acceptor pairs in 2.3–3.8 Å), hydrophobic contacts (3.2–5.6 Å), a distance-dependent electrostatic term with formal receptor charges, aromatic stacking, and a rotatable-bond entropy penalty (all in kcal mol⁻¹-like units).

Refinement. The best placement is polished by 90 greedy-or-Metropolis moves alternating random perturbations with *snap moves* that translate a ligand donor/acceptor onto the ideal hydrogen-bond distance from a complementary pocket atom. Improvements are always accepted; small regressions are accepted with a decreasing temperature to escape single-hydrogen-bond basins. Ligands for which no placement is clash-free are retained with their least-clashing pose and flagged *no-fit*; a 0.3× composite multiplier prevents steric incompatibility from being outvoted by favourable contact terms.

3.3 Quantum refinement

Classical scores treat electrons implicitly, yet binding is an electronic event: charge transfer across a hydrogen bond. For each top-ranked pose, Q-Pharm extracts the strongest polar contact—the shortest ligand donor⋯acceptor or acceptor⋯donor pair within 3.6 Å—and models it as a two-site, two-orbital Hubbard dimer with modes (1 ↑, 2 ↑, 1 ↓, 2 ↓):

$$H = -t \sum_{\sigma} \left(c_{1\sigma}^{\dagger} c_{2\sigma} + \text{h.c.} \right) + U \sum_{\sigma} n_{1\sigma} n_{1\bar{\sigma}} + \frac{\delta}{2} \sum_{\sigma} (n_{1\sigma} - n_{2\sigma}), \quad (2)$$

with distance- and chemistry-derived parameters

$$t = t_0 e^{-(d-2.9\text{Å})/0.9}, \quad \delta = 1.4 (\chi_{\text{lig}} - \chi_{\text{prot}}), \quad (3)$$

where $t_0 = 0.045 E_h$, $U = 0.28 E_h$ and χ denotes Pauling electronegativity. The four-mode Hamiltonian is mapped exactly to four qubits (Jordan–Wigner [13]; verified numerically against the fermionic matrix in the test suite) and expanded in the Pauli basis (15–16 terms after the particle-number penalty below).

The ground state is found with a VQE [10]: a hardware-efficient R_y ansatz (depth 2, twelve parameters) is optimized by L-BFGS-B with *exact parameter-shift gradients* and two random restarts. Because the ansatz does not conserve particle number, a penalty $\lambda(N-2)^2$ with $\lambda = 0.6 E_h$ pins the variational state to the physical two-electron sector—a standard technique when particle-number symmetry is not built into the circuit. The interaction energy

$$\Delta E = E_{\text{coupled}}(t) - E_{\text{decoupled}}(t=0) \quad (4)$$

quantifies charge-transfer stabilization; negative values indicate that the contact benefits from the electronic channel. All simulations run on Qiskit’s statevector backend [19]; each refined contact costs approximately four seconds on a laptop CPU.

We emphasize the intended reading of this stage: it is a *proof-of-concept refinement* on an effective model with phenomenological parameters, not an ab initio electronic-structure calculation. Its purpose is to exercise, end-to-end and openly, the architecture proposed for near-term quantum advantage in drug discovery [12, 14]; replacing the contact-parameterization step with an active-space Hamiltonian from ab initio methods would change nothing else in the loop.

3.4 ADMET profiling and machine-learning re-ranking

Each docked compound is profiled with the standard filter cascade: Lipinski Rule of 5 violations [6]; Veber oral-bioavailability rules [7]; PAINS and Brenk structural alerts via RDKit’s filter catalogue [8]; and ESOL estimated solubility [9],

$$\log S = 0.16 - 0.63 \text{ cLog}P - 0.0062 \text{ MW} + 0.066 \text{ rotB} - 0.74 \text{ AP}. \quad (5)$$

A logistic-regression drug-likeness prior over eleven RDKit descriptors is trained at build time on the drug library itself (positives) against curated non-drug chemicals—industrial solvents, dyes, pesticides and alkylating toxophores (negatives). It is a transparent illustration of the re-ranking stage rather than a production QSAR ensemble, and its predictions are displayed as such in the console.

3.5 Composite ranking

All signals are fused with fixed, published weights:

$$S = 0.55 \hat{E}_{\text{dock}} + 0.20 A + 0.15 Q + 0.10 M, \quad (6)$$

where $\hat{E}_{\text{dock}}$ is the min–max-normalized docking score (inverted so that stronger binders score higher), A the ADMET score, Q the normalized quantum stabilization and M the ML prior. Ligands flagged *no-fit* receive a 0.3× multiplier. Every candidate exposes all four components in the console.

4 Implementation

The backend is Python 3.11 with FastAPI, RDKit 2026.03, NumPy/SciPy and Qiskit 2.5 [18, 19]; the

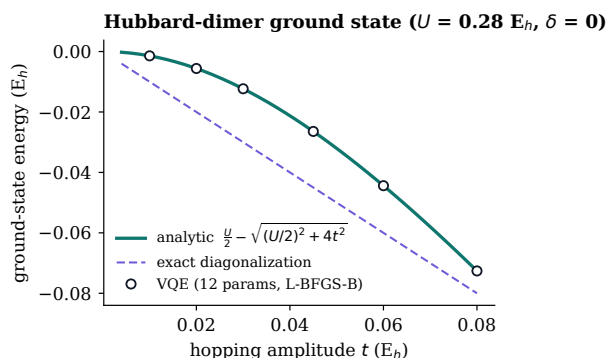


Figure 2: Hubbard-dimer ground state used by the quantum-refinement stage. The analytic half-filled solution (solid) and exact diagonalization of the four-qubit Hamiltonian (dashed) coincide; live VQE runs (circles, computed with the released implementation) land on the same curve to within $4.8 \times 10^{-12} E_h$.

frontend is React 18 with Vite, Tailwind CSS and 3Dmol.js [17]. The drug library (294 compounds spanning antiviral, antibacterial, antiparasitic, central-nervous-system, cardiovascular, oncology and anti-inflammatory classes; 38 with documented repurposing histories) is resolved from PubChem by name or CID, validated with RDKit and stored as canonical SMILES; a deterministic build script reconstructs all data artefacts and retrains the ML model. Ten curated targets were verified against RCSB at build time with explicit reference ligands (for example PJE for 6LU7, F86 for remdesivir monophosphate in the polymerase structure 7BV2, and the oseltamivir code G39 in neuraminidase 2HU4). Docked poses are exported in the original PDB coordinate frame, so poses overlay the receptor without alignment. The API exposes job submission, progress polling, structure and pose files, CSV export and an OpenAPI schema; finished screenings are addressable by permalink. The complete system is 61 tracked files, MIT-licensed.

5 Validation and results

5.1 Correctness of the quantum module

The half-filled Hubbard dimer admits the analytic ground state $E = U/2 - \sqrt{(U/2)^2 + 4t^2}$ at zero site offset. The fermionic construction reproduces this expression to $10^{-9} E_h$ across the hopping range, and the VQE converges to the same curve: across six hopping values spanning an order of magnitude, the maximum deviation between VQE and the analytic energy is $4.8 \times 10^{-12} E_h$ (Figure 2), using twelve variational parameters and two restarts. This confirms the three correctness links of the quantum chain—the fermionic Hamiltonian, its qubit mapping and the variational solver.

5.2 End-to-end screening of SARS-CoV-2 M^{pro}

We screened the default 60-drug prefiltered library against PDB 6LU7, the crystal structure of the SARS-CoV-2 main protease in complex with inhibitor N3 [3].

Pocket reproduction. The ligand-anchored pocket detector selected the co-crystallized inhibitor site and recovered 22 residues—His41 and Cys145 (the catalytic dyad), Met49, Met165, Glu166, Gln189, Asn142, Gly143 and neighbours—matching the experimentally established active site. The pocket’s chemical inventory (21 donors, 26 acceptors, 55 hydrophobic atoms, 4 aromatic rings) drives the subsequent typing and placement stages.

Screening outcome. All 60 selected drugs were docked (18,000 sampled placements; 362 s total on a consumer laptop, of which the eight VQE refinements contributed roughly 30 s). Table 1 lists the top ten by composite score; Figure 3 visualizes the same data. The top-ranked hit, amoxicillin, makes seven recorded interactions including four hydrogen bonds to Asn142, Thr26 and Leu4 at 2.66–3.11 Å, with a VQE contact stabilization of -0.50 eV at an S...N distance of 2.66 Å. Nucleoside-antiviral pharmacophores rank prominently (entecavir, sofosbuvir, molnupiravir, ribavirin), consistent with the target’s preference for peptidomimetic and polar inhibitors; the largest macromolecules (vancomycin, daptomycin) are correctly flagged *no-fit* despite locally attractive contact scores, demonstrating that the steric guard behaves as designed.

Negative controls. Molecules that physically cannot occupy the pocket are handled transparently: vancomycin (101 heavy atoms) and daptomycin (115) return their least-clashing poses with unfavourable net scores and the *no-fit* flag, and are excluded from quantum refinement. In an earlier build without full-protein clash checking, such compounds could rank spuriously high; the guard was added and is covered by regression tests.

5.3 Performance

On a consumer laptop (Windows 11, no GPU), the 60-drug screen completes in 362 ± 10 s: pocket detection and prefiltering take under 3 s, docking averages ~ 5 s per drug (dominated by conformer generation for macromolecules), and each VQE refinement adds ~ 4 s. The Fast preset (25 drugs) completes in roughly 2.5–3 minutes. Memory use is modest; the system runs entirely from simulators and public databases with no proprietary dependencies.

6 Limitations and discussion

Docking realism. The receptor is rigid, waters and protonation states are not modelled, and side-chain flexibility is absent. The scoring function is empirical and calibrated only against pocket geometry and known complex chemistry; absolute scores should be

Table 1: Top ten candidates from the 60-drug screening of SARS-CoV-2 M^{pro} (PDB 6LU7). ΔG is the empirical docking score of Equation 1; ΔE is the VQE interaction energy of the strongest polar contact (dash: pose not among the quantum-refined top eight); A is the ADMET grade (A best to D worst); S is the composite of Equation 6. Values are computational estimates, not measurements.

Rank	Compound	Class	ΔG (kcal/mol)	ΔE (eV)	A	S	Contacts of best pose
1	Amoxicillin	β -lactam antibiotic	-12.9	-0.50	A	0.700	4 H-bonds (Asn142, Thr26)
2	Mannitol	Osmotic diuretic	-8.6	-1.07	A	0.670	polyol H-bond network
3	Zanamivir	Neuraminidase inhib.	-8.6	-1.57	D	0.639	glycerol-chain H-bonds
4	Sulfasalazine	Aminosalicylate	-14.6	-0.51	C	0.620	sulfonamide + azo contacts
5	Entecavir	Nucleoside analog (HBV)	-7.6	-0.44	A	0.572	exocyclic amine H-bonds
6	Ceftriaxone	Cephalosporin	-9.8	-1.03	D	0.558	O14...N 2.66 Å contact
7	Sofosbuvir	NS5B inhibitor (HCV)	-8.2	-0.61	C	0.535	phosphoramidate contacts
8	Molnupiravir	Nucleoside analog	-7.5	—	B	0.529	cytidine-analog H-bonds
9	Ribavirin	Broad-spectrum antiviral	-5.5	—	A	0.525	base-mimic H-bonds
10	Dabigatran	Thrombin inhibitor	-8.6	-0.44	C	0.518	benzimidazole stacking

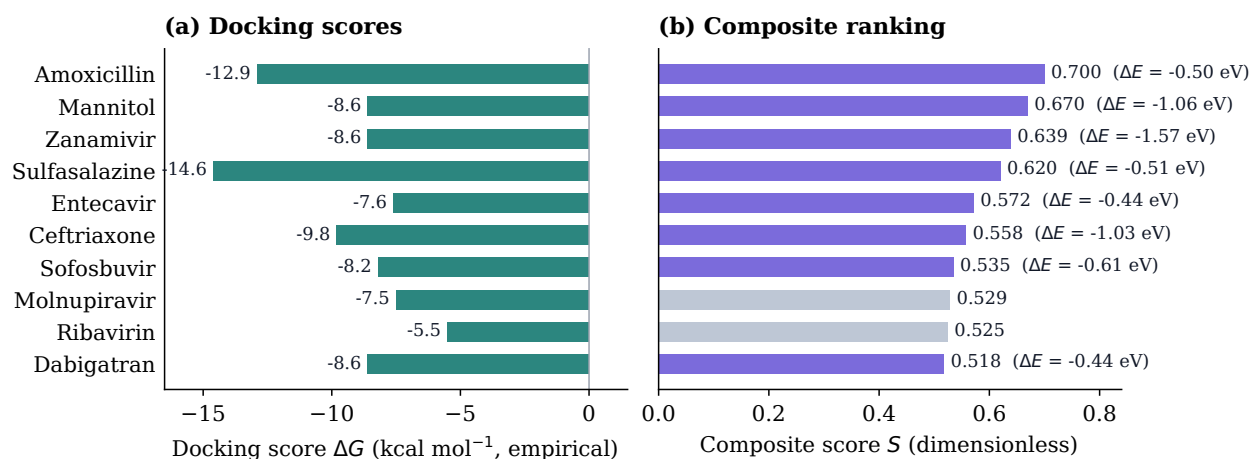


Figure 3: Real output of the released pipeline for the 60-drug screen of SARS-CoV-2 M^{pro} (PDB 6LU7). (a) Empirical docking scores of the top ten candidates by composite rank; more negative is better. (b) Composite scores; violet bars indicate candidates whose strongest contact was refined by the four-qubit VQE (with the resulting charge-transfer stabilization ΔE), grey bars were not among the quantum-refined top eight.

compared *within* a run, never interpreted as free energies.

Quantum layer. The Hubbard-dimer refinement uses phenomenological parameters and a statevector simulator. It demonstrates the hybrid architecture honestly and validates it numerically, but it does not produce ab initio energetics. We consider this the correct scope for an open reference implementation; the API contract isolates the parameterization so that an active-space Hamiltonian can replace it without touching the pipeline.

ADMET. The ML prior is trained on a small curated set and saturates for very drug-like molecules; it discriminates gross non-drug chemistry reliably but should not be read as a pharmacokinetics prediction.

Library coverage. 294 curated compounds is a deliberate, license-clean subset, not the full DrugBank; the build script is designed for extension.

Intended use. Q-Pharm ranks hypotheses. Any candidate of interest requires literature follow-up and experimental validation before anyone should draw biological or medical conclusions from it.

7 Conclusion and availability

Q-Pharm shows that a meaningful drug-repurposing screen—structure acquisition, pocket detection, guided docking, a four-qubit VQE refinement, ADMET filtering and an interpretable composite ranking—can now be assembled entirely from open components and run on a laptop in minutes. The quantum stage is implemented as real, validated hybrid quantum-classical computation on an effective model, providing a reproducible scaffold that ab initio methods can upgrade in place. The system is available under the MIT license at <https://github.com/Shreyasderaje/q-pharm>, with a browser console, documented API, deterministic data builds and a test suite covering every module, including the analytic correctness of the quantum chain.

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References

- [1] Pushpakom, S. et al. Drug repurposing: progress, challenges and recommendations. *Nat. Rev. Drug Discov.* **18**, 41–58 (2019).
- [2] Beigel, J. H. et al. Remdesivir for the treatment of Covid-19 — final report. *N. Engl. J. Med.* **383**, 1813–1826 (2020).
- [3] Jin, Z. et al. Structure of M^{Pro} from SARS-CoV-2 and discovery of its inhibitors. *Nature* **582**, 289–293 (2020).
- [4] Yin, W. et al. Structural basis for inhibition of the RNA-dependent RNA polymerase from SARS-CoV-2 by remdesivir. *Science* **368**, 1499–1504 (2020).
- [5] Böhm, H.-J. The computer program LUDI: a new method for the de novo design of enzyme inhibitors. *J. Comput. Aided Mol. Des.* **6**, 61–78 (1992).
- [6] Lipinski, C. A., Lombardo, F., Dominy, B. W. & Feeney, P. J. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Adv. Drug Deliv. Rev.* **23**, 3–25 (1997).
- [7] Veber, D. F. et al. Molecular properties that influence the oral bioavailability of drug candidates. *J. Med. Chem.* **45**, 2615–2623 (2002).
- [8] Baell, J. B. & Holloway, G. A. New substructure filters for removal of pan assay interference compounds (PAINS) from screening libraries. *J. Med. Chem.* **53**, 2719–2740 (2010).
- [9] Delaney, J. S. ESOL: estimating aqueous solubility directly from molecular structure. *J. Chem. Inf. Comput. Sci.* **44**, 1000–1005 (2004).
- [10] Peruzzo, A. et al. A variational eigenvalue solver on a photonic quantum processor. *Nat. Commun.* **5**, 4213 (2014).
- [11] Kandala, A. et al. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature* **549**, 242–246 (2017).
- [12] Cao, Y., Romero, J., McClean, J. P., Hastings, E., Babbush, R. & Aspuru-Guzik, A. Quantum chemistry in the age of quantum computing. *Chem. Rev.* **119**, 10856–10915 (2019).
- [13] Jordan, P. & Wigner, E. Über das Paulische Äquivalenzverbot. *Z. Phys.* **47**, 631–651 (1928).
- [14] Li, J. et al. A hybrid quantum computing pipeline for real world drug discovery. *Sci. Rep.* **14** (2024).
- [15] Berman, H. M. et al. The Protein Data Bank. *Nucleic Acids Res.* **28**, 235–242 (2000).
- [16] Kim, S. et al. PubChem 2023 update. *Nucleic Acids Res.* **51**, D523–D531 (2023).
- [17] Rego, N. & Koes, D. 3Dmol.js: molecular visualization with WebGL. *Bioinformatics* **31**, 1322–1324 (2015).
- [18] Landrum, G. et al. RDKit: open-source cheminformatics software. <https://www.rdkit.org> (accessed October 2026).
- [19] Javadi-Abhari, A. et al. Quantum computing with Qiskit. *arXiv:2405.08810* (2024).